

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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MODEL PRACTICAL EXAMINATION – SET 5

*Answer all questions. Each question carries equal marks (20).
marks: 100*

Total

1. Create a simple cloud software application for Flight Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as one way, round trip, From, To, Departure, Travelers & Class, Fare Type, International and Domestic Flights, etc.
2. Create a simple cloud software application for Property Buying & Rental process (in Chennai city) using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Buy, Rent, Commercial, Rental Agreement, Area, Place, Range, Property Value, etc.
3. Create a Snapshot of a VM and test it by loading the previous version/cloned VM.
4. Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of your present Windows environment.

1. Create a simple cloud software application for Flight Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as one way, round trip, From, To, Departure, Travelers & Class, Fare Type, International and Domestic Flights, etc.

Aim

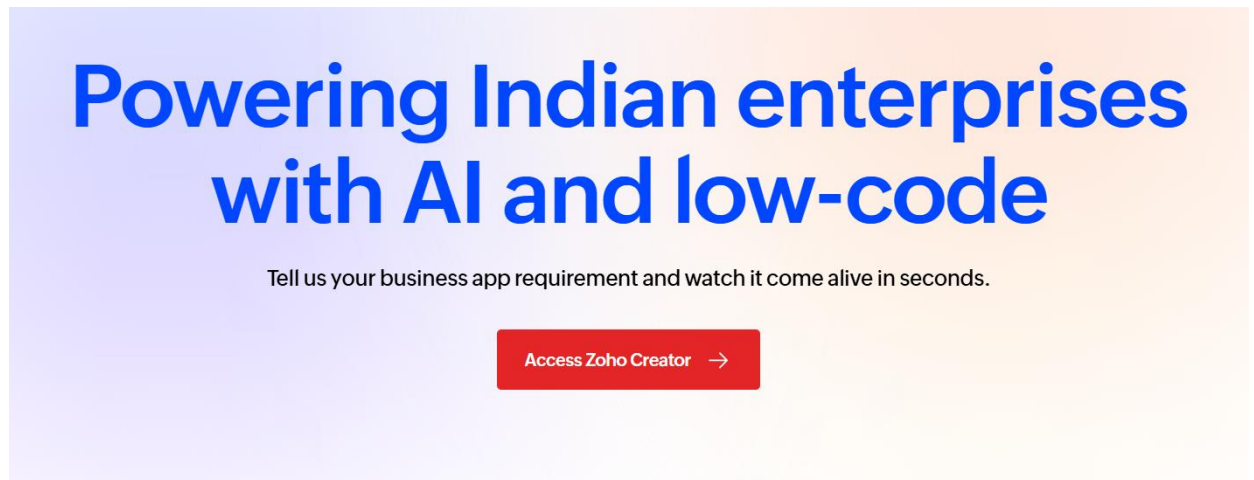
To create a simple Flight Reservation System using Google Forms to demonstrate the Software as a Service (SaaS) model.

Algorithm

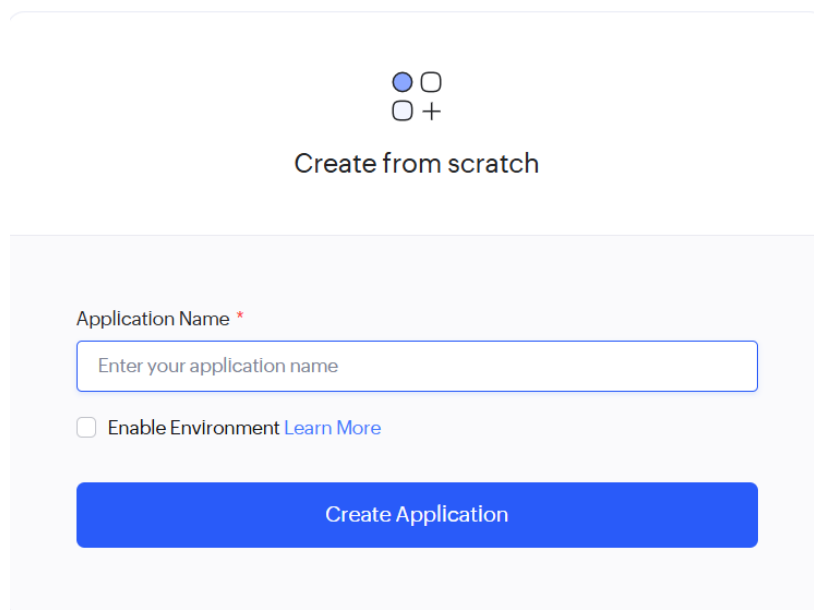
1. Sign in to your Google account and open Google Drive.
2. Click New → Google Forms → Blank Form and enter the title Flight Reservation System.
3. Add fields such as Passenger Name, Email, Mobile Number, From, To, Departure Date, Return Date, Travelers, Class, Fare Type, Domestic/International, and Trip Type (One Way/Round Trip).
4. Set important fields like From, To, and Departure Date as required.
5. Enable Collect Email Addresses from the Settings menu.
6. Click Send, generate the shareable form link, and submit a sample booking.
7. Verify that the submitted booking details are stored automatically in the Responses section.

IMPLEMENTATION

Step 1: Sign in to your Google account and open Google Drive.



Step 2: Create a new application from scratch using Google Forms.

A screenshot of the "Create from scratch" form in Zoho Creator. The form is light gray with rounded corners. At the top, there are two rows of icons: the first row has a blue circle and an empty circle, and the second row has an empty circle and a plus sign. Below the icons is the text "Create from scratch". Further down, there is a label "Application Name *" followed by a text input field containing the placeholder "Enter your application name". Below the input field is a checkbox labeled "Enable Environment" with a link "Learn More" to its right. At the bottom of the form is a large blue button with the text "Create Application".

Step 3: Click New → From scratch Google Forms → Blank Form and enter the required form

How would you like to create your form?

On my own

+

From scratch

Import with data

Using Zia

Step 4: Create the Airline Form and add fields such as Airline Name, Flight Number, From, To, Flight Type (Domestic/International), Fare Type, and Class.

Flight Ticket Reservation System

FTR Flight Ticket Reservation System

Airlines Form

Airlines Form

Airlines Form Report

Passengers form

Reservation form

Dashboard

Airlines Form

Airlines

First NameLast Name

AirlineName

Country

Submit

Flight Ticket Reservation System

FTR Flight Ticket Reservation System

Airlines Form Report

Airlines Form Report

Passengers form

Reservation form

Dashboard

Airlines	Airlines	AirlineName	Country
AL007	7	Lufthansa	Germany
AL006	6	British Airways	British Airways
AL005	5	Qatar Airways	Qatar
AL004	4	singapore Airlines	singapore
AL003	3	emirates	IUSA
AL002	2	IndiGo	india
AL001	1	Air India	India

Step 5: Create the Passenger Form and add fields such as Passenger Name, Email, Mobile Number, Age, and Gender.

Flight Ticket Reservation System

FTR

Flight Ticket Reservation System

Airlines Form

Passengers form

Passengers form

Passengers form Report

Reservation form

Dashboard

Passengers form

Passenger

First Name

Last Name

Name

Email

Phone

#####

Submit

Flight Ticket Reservation System

Trial expires in 7 days Upgrade Edit this application Help

FTR

Flight Ticket Reservation System

Airlines Form

Passengers form

Passengers form

Passengers form Report

Reservation form

Dashboard

Passengers form Report

	Passenger	Name	Email	Phone	PassengerID
	P007	vikram	vikram@gmail.com	7695841230	7
	P006	meena	meena@gmail.com	8695231470	6
	P005	karthik	karthik@gmail.com	2365417890	5
	P004	sneha reddy	sneha@gmail.com	3214569870	4
	P003	arjun kumar	arjun@gmail.com	3214569870	3
	P002	priya nair	priya@gmail.com	9874521036	2
	P001	rahul	rahul@gmail.com	9874563210	1

Step 6: Create the Reservation Form with Trip Type, Departure Date, Return Date, Travelers, Airline, Passenger, and Seat Class fields.

Flight Ticket Reservation System

FTR

Flight Ticket Reservation System

Airlines Form

Passengers form

Reservation form

Reservation form

Reservation form Report

Dashboard

Reservation form

Passenger

First Name

Last Name

Flight number

#####

booking date

dd-MMM-yyyy

Submit

Flight Ticket Reservation System

FTR

Flight Ticket Reservation System

Airlines Form

Passengers form

Reservation form

Reservation form

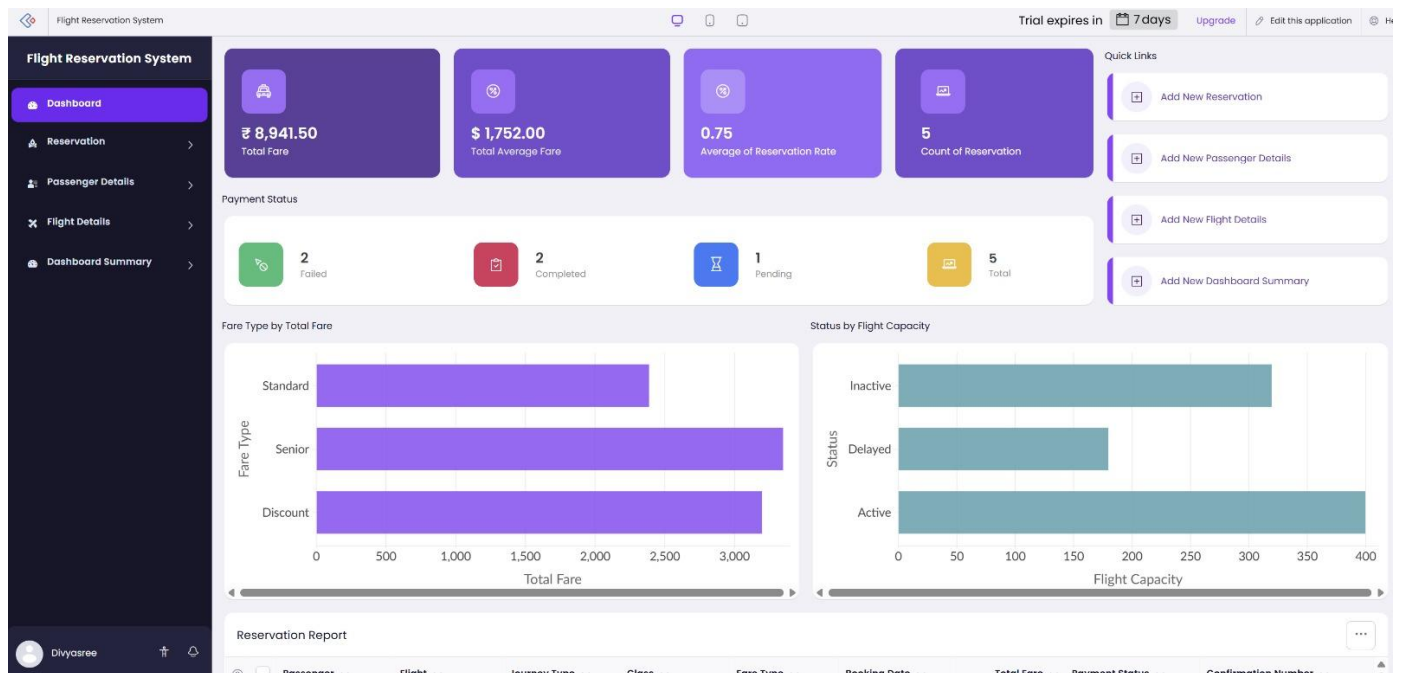
Reservation form Report

Dashboard

Reservation form Report

☐	Passenger	reservationId	Flight number	booking date
☐	Anjali Gupta	8	56708	03-Jul-2026
☐	Vikram Singh	7	23456	18-Jul-2026
☐	Meena Devi	6	24692	31-Jul-2026
☐	Karthik Raj	5	86432	23-Jul-2026
☐	Sneha Reddy	4	98775	10-Jul-2026
☐	Arjun Kumar	3	76544	01-Jul-2026
☐	Priya Nair	2	56794	17-Jul-2026
☐	ragul	1	65231	28-Jun-2026

Step 7: Verify that the submitted booking details are stored automatically in the Responses section.



Result

A cloud-based Flight Reservation System was successfully created using Google Forms, and booking details were stored automatically in the cloud.

2.Create a simple cloud software application for Property Buying & Rental process (in Chennai city) using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Buy, Rent, Commercial, Rental Agreement, Area, Place, Range, Property Value, etc.

Aim

To create a cloud-based Property Buying & Rental System for Chennai using Google Forms to demonstrate SaaS.

Algorithm

Step 1: Sign in to Zoho and create a new Application Form.

Step 2: Enter the title Chennai Property Buying & Rental System.

Step 3: Add fields such as Customer Name, Buy/Rent, Residential/Commercial, Rental Agreement, Area, Preferred Place, Budget Range, and Property Value.

Step 4: Add Chennai locations such as T. Nagar, Anna Nagar, Velachery, Adyar, OMR, and Tambaram as dropdown options.

Step 5: Mark essential fields like Requirement, Area, Budget Range, and Property Value as mandatory.

Step 6: Click Send, generate the form link, and submit a sample property request.

Step 7: Check the Responses section to confirm that all property details are stored automatically.

IMPLEMENTATION

Step 1: Sign in to your Google account and Access zoho creator

Powering Indian enterprises with AI and low-code

Tell us your business app requirement and watch it come alive in seconds.

Access Zoho Creator →

Step 2: Create a new application from scratch using Google Forms.



Create from scratch

Application Name *

Enter your application name

☐ Enable Environment [Learn More](#)

Create Application

Step 3: Click New → From scratch Google Forms → Blank Form and enter the required form

How would you like to create your form?

On my own



From scratch



Import with data



Using Zia

Step4: Create the Customer Registration Form with customer details.

Property Buying & Rental System

Customer Registration

Customer Name: sai

Phone: +91 08309018910

Email: sai341021@gmail.com

Address: Thandalam

Customer Type: Tenant

Submit

Property Buying & Rental System

Customer Registration Report

Customer ID	Customer Name	Phone	Email	Address	Customer Type
2	Tarun	+919632587410	tarun@gmail.com	Flat no:502, vigneshwara heights apartment	Tenant
1	Mrudula	+919874563210	sai@gmail.com	saveetha nagar, saveetha university	Buyer
3	sai	+918309018910	sai341021@gmail.com	Thandalam	Buyer

Step 5: Create the **Rental Agreement Form** with rental terms and agreement details.

Property Buying & Rental System

Booking & Rental Agreement

Customer: -Select-

Property: -Select-

Agreement Type: -Select-

Agreement Date: dd-MMM-yyyy

Start Date: dd-MMM-yyyy

End Date: dd-MMM-yyyy

Advance Amount: ##.##.###.##

Status: -Select-

Submit

https://creatorapp.zoho.in/saimrudulaa2087.sse_savee/property-buying-rental-system#Form:Booking_Rental_Agreement

Property Buying & Rental System

Booking & Rental Agreement Report

Booking ID	Customer	Property	Agreement Type	Agreement Date	Start Date	End Date
1	Mrudula	House	Sale	24-Jul-2026	25-Jul-2026	25-Sep-2026

Showing 1 of 1

Step 6: Create the Property Details Form with Area, Place, Range, and Property Value fields.

Property Buying & Rental System

Property details

Property Type:

Property Name:

Area:

Location:

Property Value:

Rental Amount:

Availability:

Range:

Bedrooms:

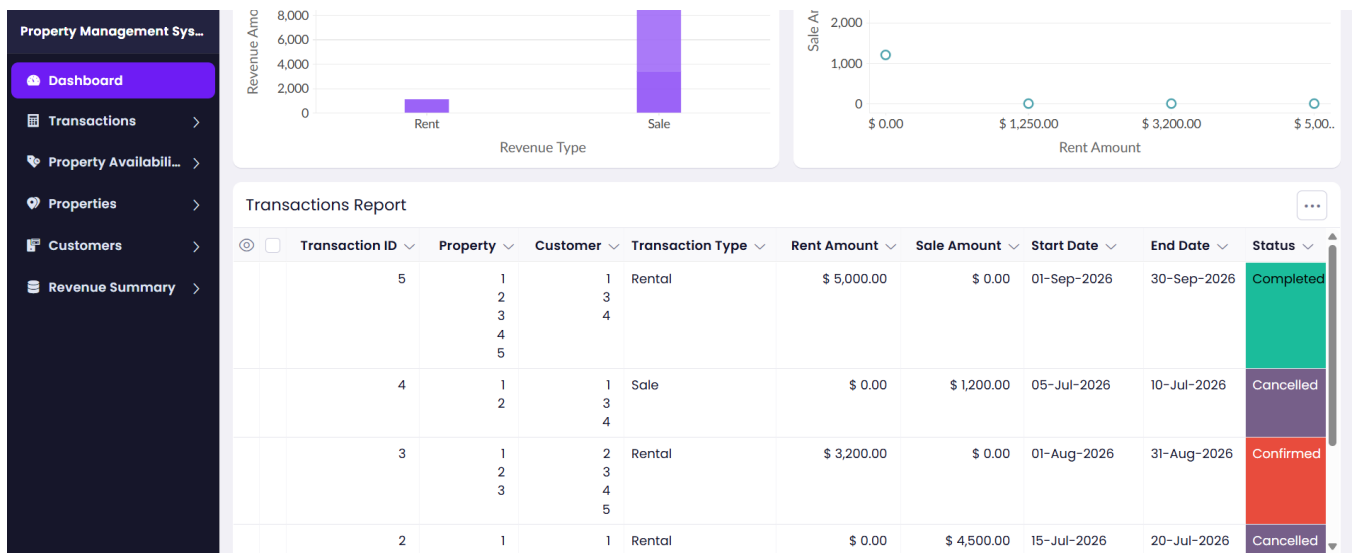
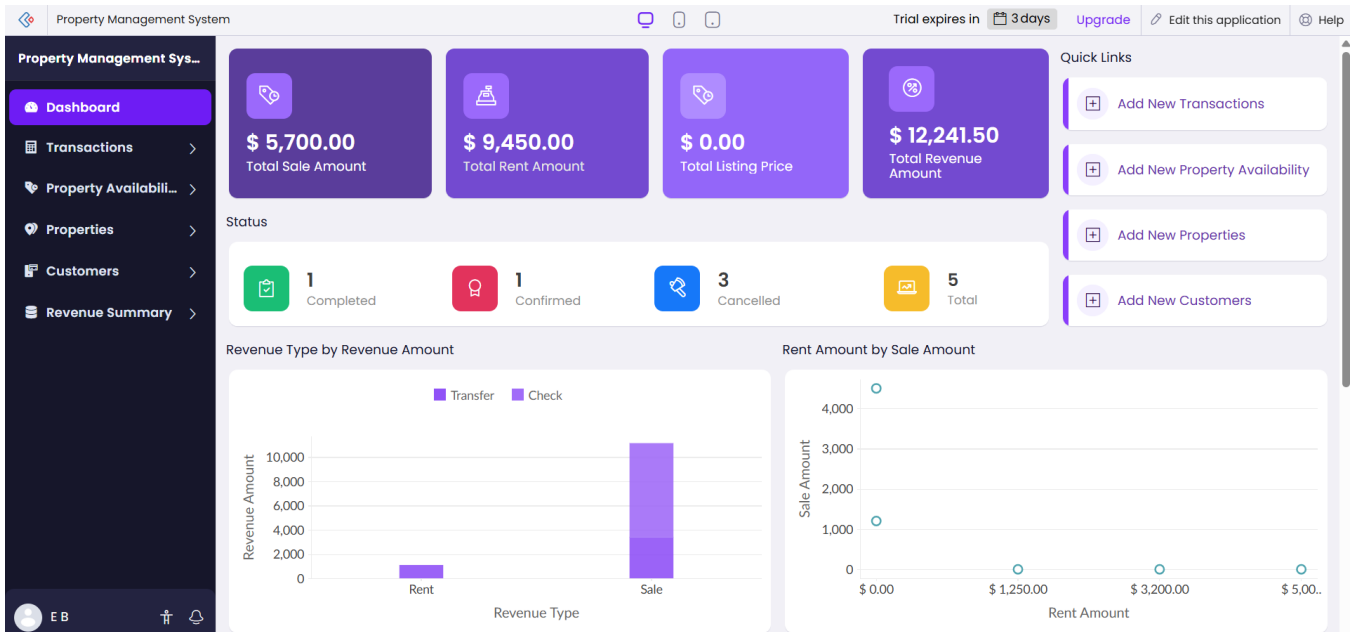
Bathrooms:

Property Buying & Rental System

Property details Report

Property ID	Property Type	Property Name	Area	Location	Property Value	Rental Amount
1	Buy	House	1200	Choice 1	₹ 10,000.00	₹ 5,000.00

Step 7: Check the Responses section to confirm that all property details are stored automatically.



3.Create a Snapshot of a VM and test it by loading the previous version/cloned VM.

Aim

To create a virtual machine snapshot and restore the VM to its previous saved state using VMware Workstation.

Procedure

Step 1: Open VMware Workstation and select the Ubuntu virtual machine.

Step 2: Power on the virtual machine and log in to the Ubuntu desktop.

Step 3: Click VM → Snapshot → Take Snapshot, enter the snapshot name Before_Test, and save it.

Step 4: Create a new folder or text file inside the virtual machine to make changes.

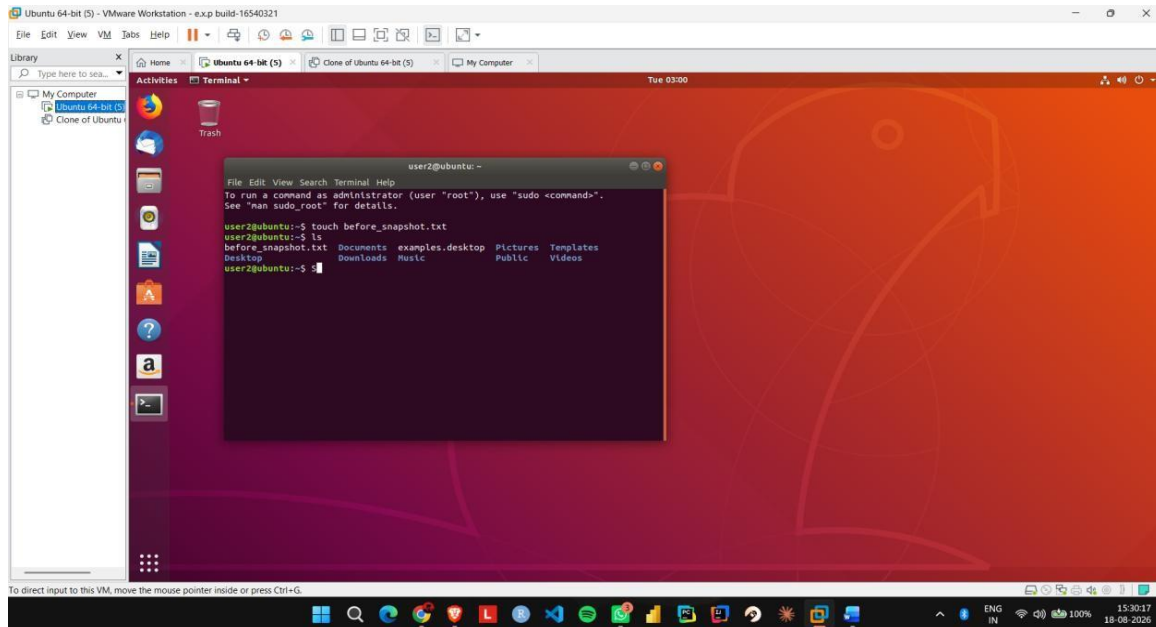
Step 5: Open VM → Snapshot → Snapshot Manager and select the saved snapshot.

Step 6: Click Go To Snapshot and confirm the restoration process.

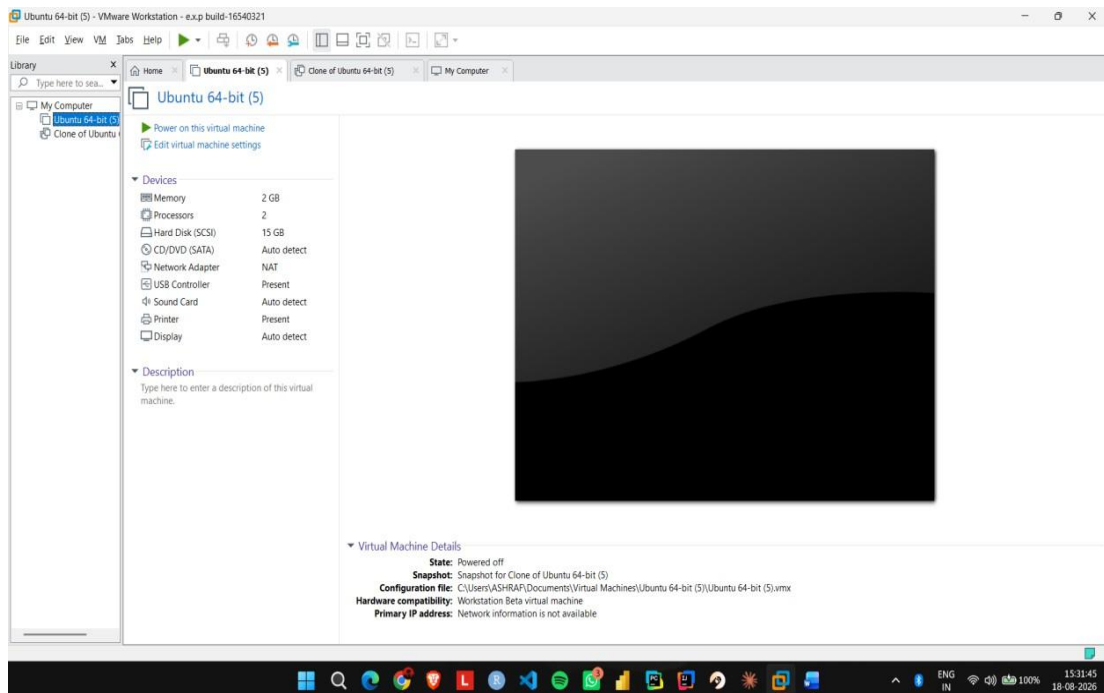
Step 7: Verify that the recently created folder or file has disappeared, confirming that the previous VM state has been restored.

IMPLEMENTATION

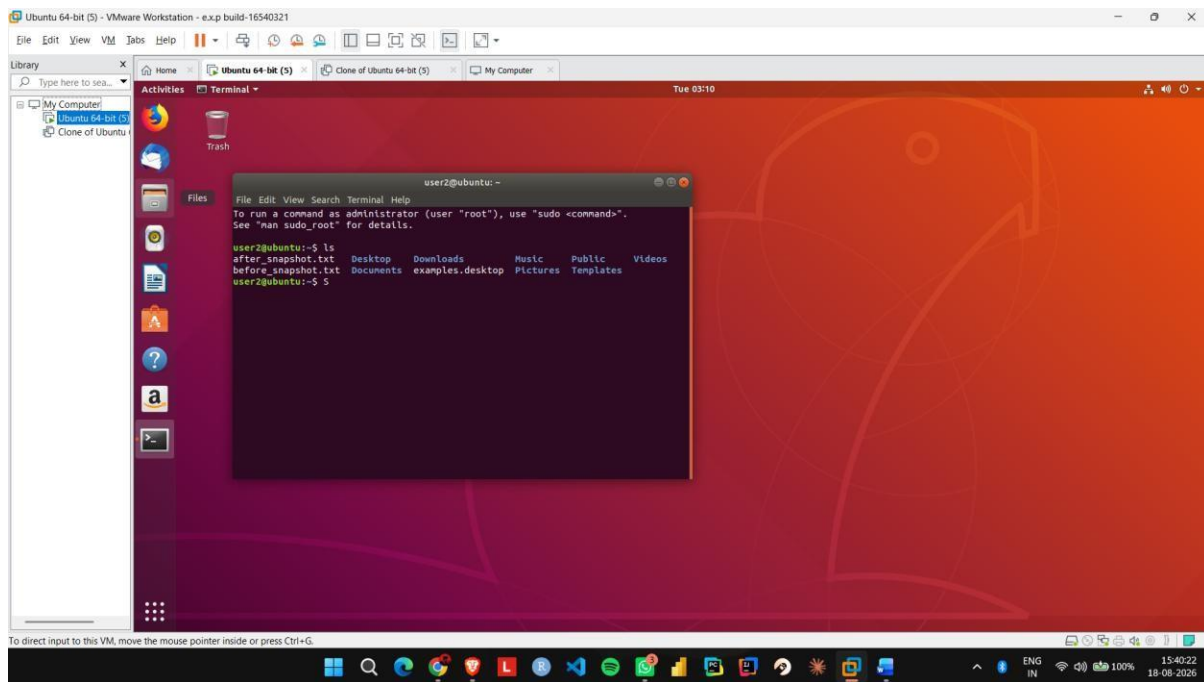
Step 1: Start the Ubuntu VM in VMware Workstation and create a test file or folder.



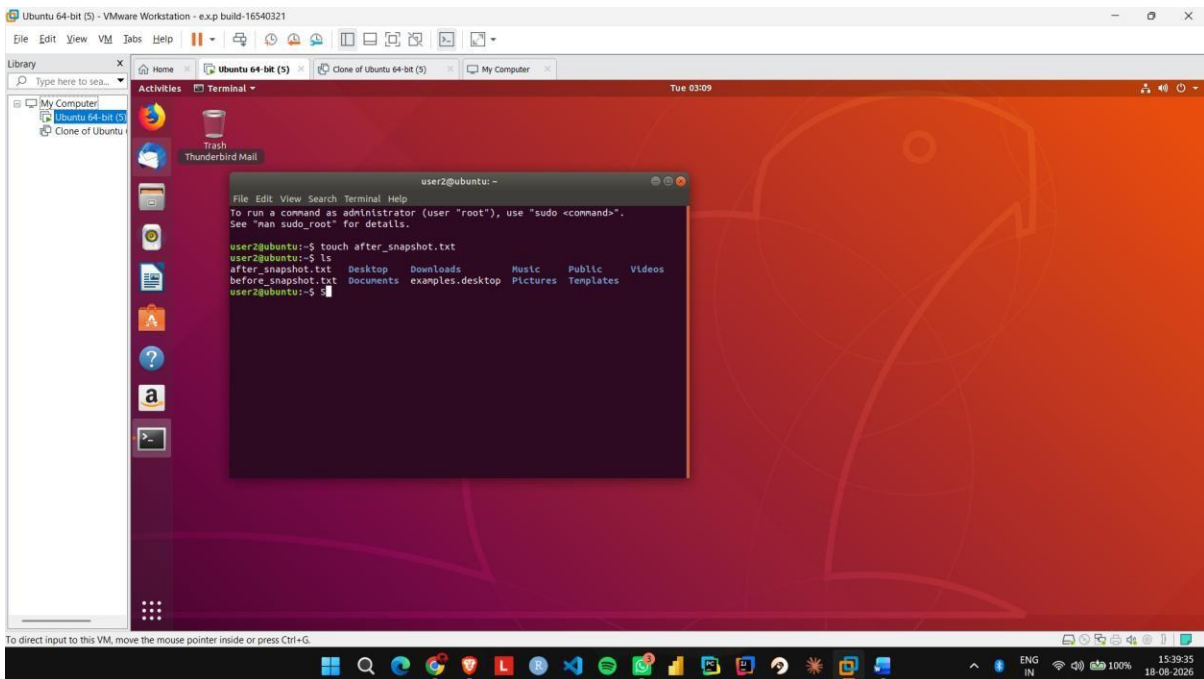
Step 2: Shut down or suspend the VM and open the Snapshot option.



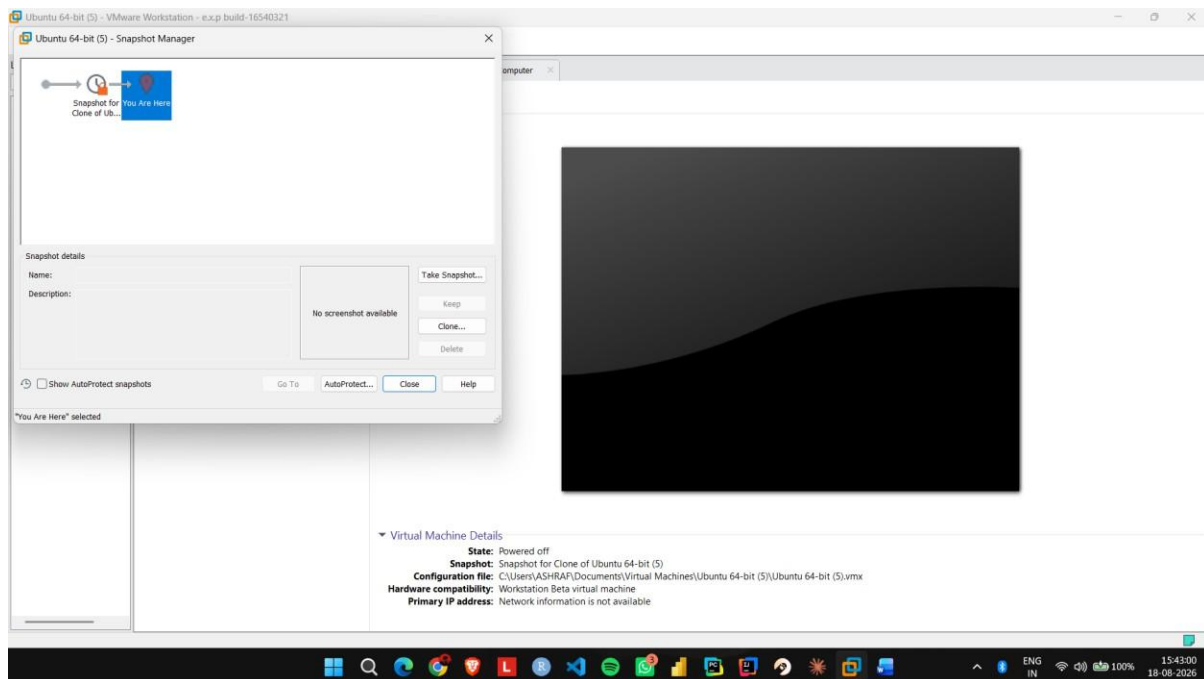
Step 3: Select Take Snapshot and enter a name such as Before Changes.



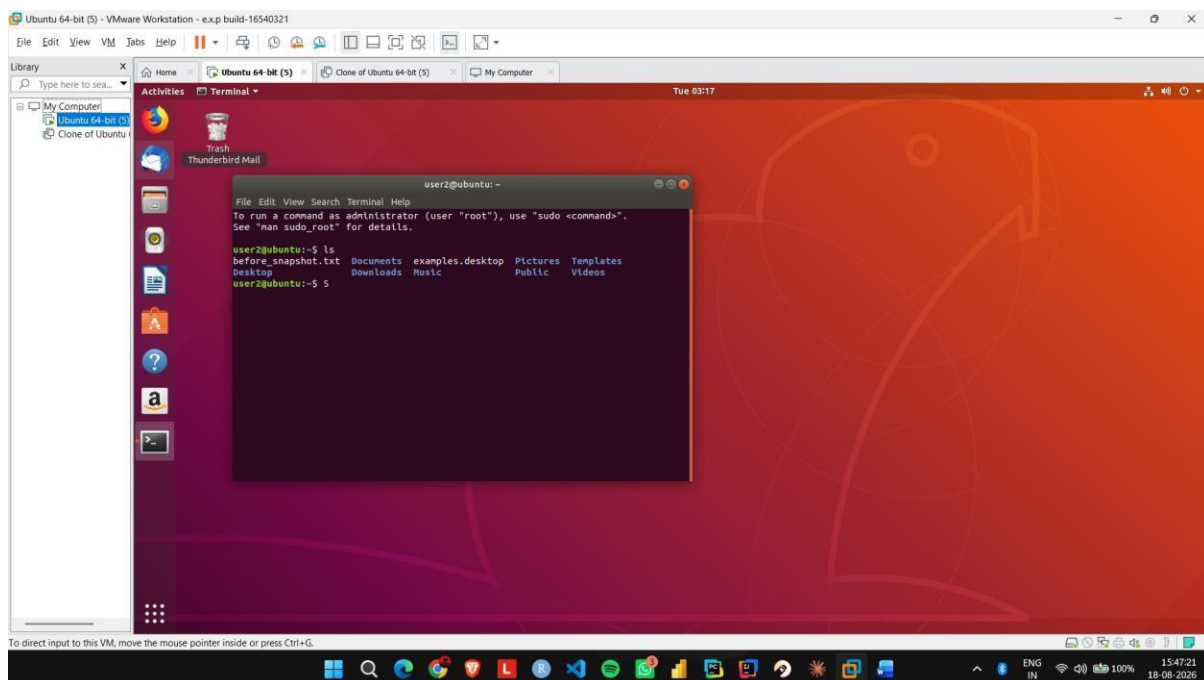
Step 4: Start the VM again and make some changes, such as creating a new file.



Step 5: Open Snapshot Manager, select the Before_Changes snapshot, and click Go To/Restore.



Step 6: Start the VM and verify that the changes made after the snapshot have been removed.



Result

The virtual machine snapshot was successfully created and restored to its previous working state.

4.Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of your present Windows environment.

Aim

To install VMware Workstation on Windows and install Ubuntu Linux as a guest operating system.

Procedure

Step 1: Download and install **VMware Workstation Pro** on the Windows computer.

Step 2: Download the **Ubuntu Desktop ISO** file from the official Ubuntu website.

Step 3: Open VMware and click **Create a New Virtual Machine**.

Step 4: Select the Ubuntu ISO file and configure the virtual machine with suitable RAM, CPU, and disk space.

Step 5: Start the virtual machine and begin the Ubuntu installation process.

Step 6: Complete the installation by selecting the language, keyboard layout, username, and password, then restart the virtual machine.

Step 7: Log in to Ubuntu and verify that the operating system is installed and functioning correctly.

IMPLEMENTATION

STEP 1: CREATING VM AND PROVIDING ISO FILE

Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Name: ✓

Folder:

ISO Image:

Edition:

Type: 64

Version:

☐ Skip Unattended Installation

Detected OS type: Ubuntu (64-bit). This OS type can be installed unattendedly. The install will start after this wizard is closed.

Help Expert Mode Back Next Cancel

STEP 2: FIG :PROVIDING CREDENTIALS FOR THE VM

Unattended Guest OS Install Setup

You can configure the unattended guest OS install by modifying username, password, and hostname. Additionally you can enable guest additions install. For Microsoft Windows guests it is possible to provide a product key.

Username and Password

Username: ✓

Password:

Repeat Password:

Additional Options

Product Key:

Hostname: ✓

Domain Name:

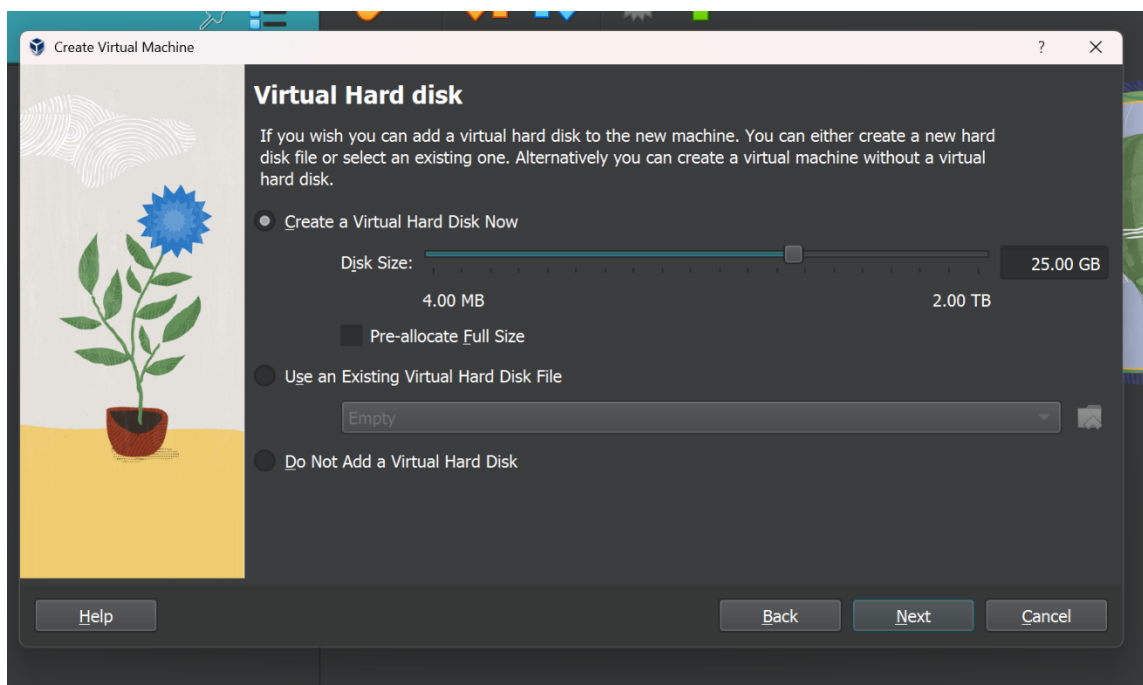
☐ Install in Background

Guest Additions

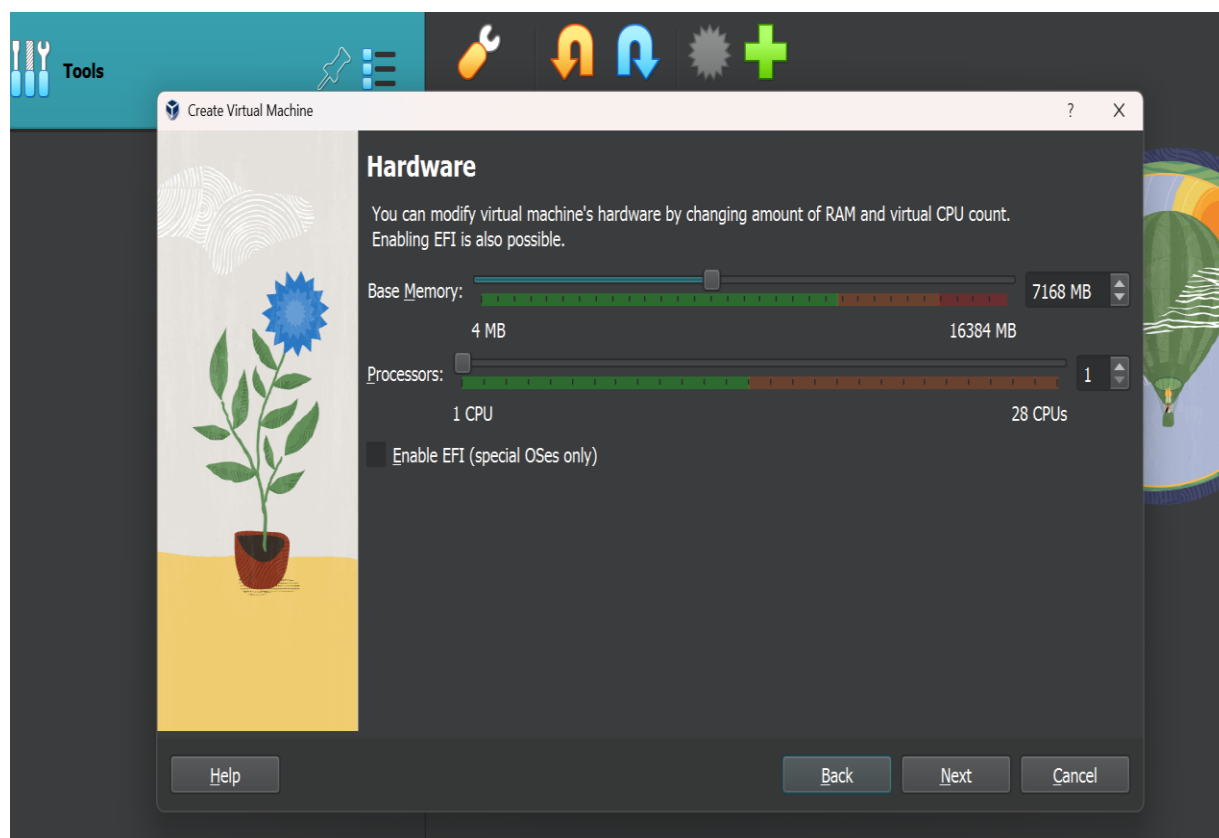
Guest Additions ISO:

Help Back Next Cancel

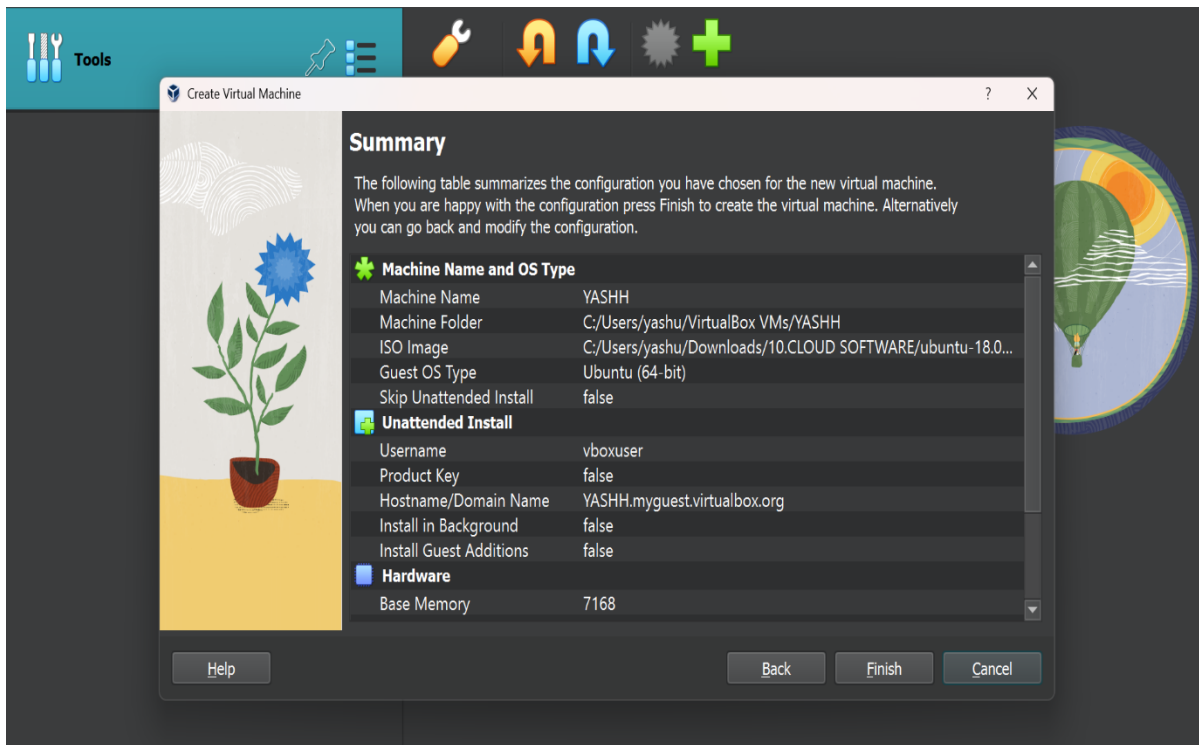
STEP



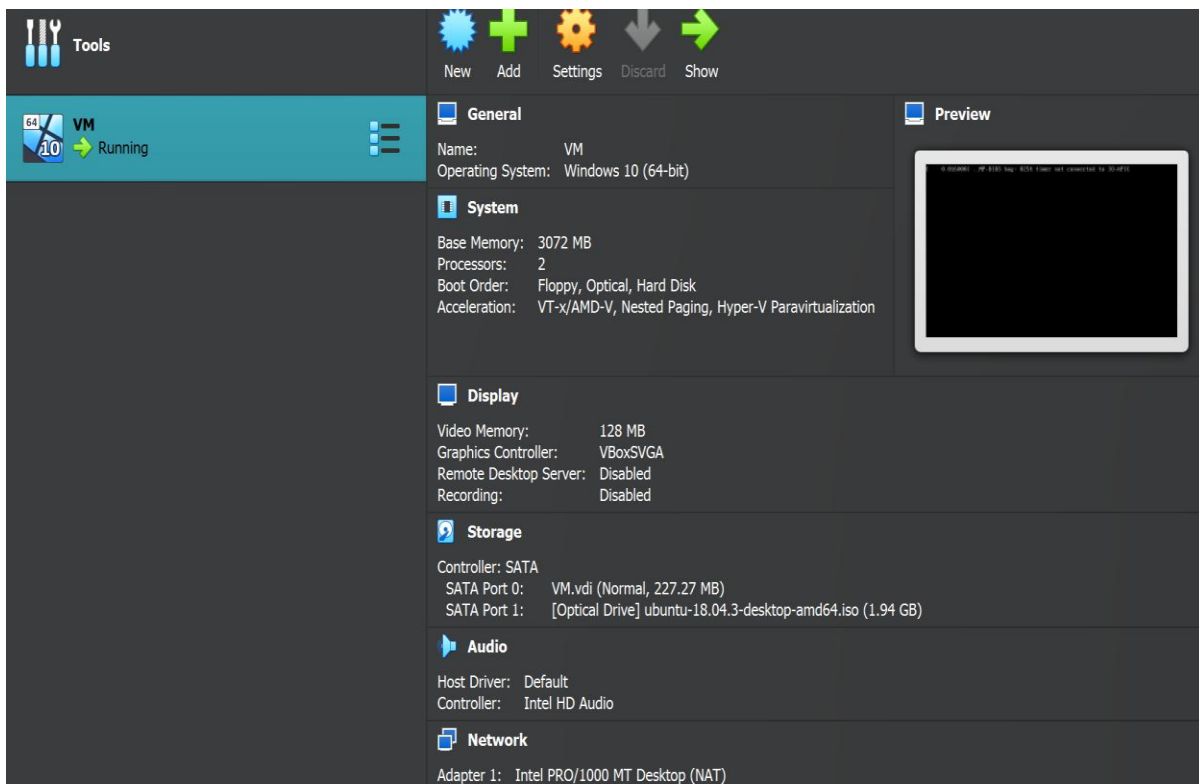
STEP3: CREATING VIRTUAL HARD DISK WITH REQUIRED CONFIGURATION



STEP 4: CONFIGURING THE VM



STEP 5:LAST STEP OF VM



STEP 6:INSTALLATION OF VM IS FINISHED

Result

VMware Workstation and Ubuntu Linux were successfully installed, and the virtual machine operated correctly within the Windows environment.